Components List



1. Arduino Uno R3 Compatible Board

WHY IT IS USED

The main microcontroller board (the brain of the project). It runs your code, reads the sensors and controls motors, displays and other outputs.



2. TowerPro SG90 9G Micro Servo Motor (180°)

WHY IT IS USED

A small motor that rotates to a precise angle between 0° and 180°. Used for robotic arms, gates, flaps and steering.



3. Raindrop Detection Sensor Module

WHY IT IS USED

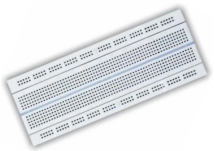
Detects water drops on its sensing plate. Used for rain alarms, automatic wipers and window/cover closing systems.



4. IR (Infrared) Obstacle Avoidance Sensor Module

WHY IT IS USED

Sends out infrared light and detects its reflection to sense nearby objects. Used for obstacle-avoiding robots, line followers and object counters.



5. GL12 840 Points Solderless Breadboard

WHY IT IS USED

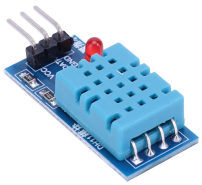
A reusable board for building circuits without soldering. Used to connect components quickly and prototype projects.



6. HC-SR04 Ultrasonic Distance Sensor Module

WHY IT IS USED

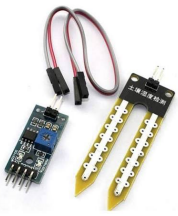
Measures distance using ultrasonic sound waves. Used for distance measurement, parking sensors, obstacle detection and water-level monitoring.



7. DHT11 Humidity & Temperature Sensor Module

WHY IT IS USED

Measures air temperature and relative humidity. Used in weather stations, room monitors and smart home projects.



8. Soil Moisture Sensor Module

WHY IT IS USED

Measures how wet or dry the soil is. Used in automatic plant watering and smart irrigation systems.



9. Heat Flame Sensor

WHY IT IS USED

Detects fire or flame through its infrared sensitivity. Used in fire alarm and fire-safety projects.



10. TTP223 Touch Switch Module

WHY IT IS USED

A capacitive touch switch that works with a finger touch instead of a physical button. Used for touch-controlled lights and switches.



11. 0.96 Inch OLED Display (128x64, Blue)

WHY IT IS USED

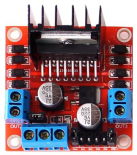
A small screen that shows text, numbers and simple graphics. Used to display sensor readings, messages and status.



12. Jumper Wires

WHY IT IS USED

Flexible wires with connector ends. Used to connect the Arduino, sensors and breadboard together.



13. Motor Driver L298N (2A Dual, PWM Control)

WHY IT IS USED

Lets the Arduino control the speed and direction of DC motors, which it cannot power directly. Used in robot cars and motorised projects.



14. DC Motor (Gear Motor)

WHY IT IS USED

A motor that spins continuously. Used to drive the wheels of robot cars and other moving parts.



15. PCB Mounted Passive Buzzer Module

WHY IT IS USED

Produces beeps and tones when driven by a signal. Used for alarms, alerts and playing simple melodies.



16. LED Traffic Light Module 5V

WHY IT IS USED

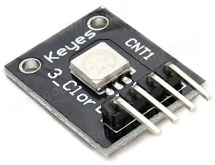
Red, yellow and green LEDs on one board. Used to build traffic light simulations and status indicators.



17. 74HC595 3-Digit Display Module

WHY IT IS USED

A 3-digit seven-segment display driven by a 74HC595 chip. Used to show numbers such as counters, temperature or timers.



18. RGB LED Module

WHY IT IS USED

An LED that mixes red, green and blue to produce many colours. Used for mood lighting, colour indicators and light effects.



19. DS18B20 Digital Temperature Sensor Module

WHY IT IS USED

A digital sensor giving accurate temperature readings. Used in thermometers, temperature monitoring and control systems.